

Introduction



- ❖ The Manual Screw Press Oil Expeller is a compact and efficient .
- ❖ It uses a simple screw-press mechanism to extract oil without electricity.
- ❖ With a capacity of up to 0.7 liters per batch, it offers a practical and convenient solution.
- ❖ This project demonstrates a simple, low-cost approach to efficient oil extraction.

Purposes of Manual Screw Press Oil Expeller

- ❖ To make oil extraction more accessible for MSMEs.
- ❖ To reduce the initial cost of starting an oil-processing business.
- ❖ To support local entrepreneurs in producing value-added oil products.
- ❖ To promote a simple, practical, and sustainable production method.
- ❖ To create new opportunities for MSMEs through small-scale oil processing.

Working Principle of Manual Screw Press Oil Expeller

- ❖ Oilseeds are fed into the hopper at the top of the machine.
- ❖ The hand-operated flywheel is rotated using the attached handle.
- ❖ The flywheel transfers motion through the gear system.
- ❖ The gear system drives the main shaft and screw press to compress the seeds.
- ❖ The pressure extracts the oil, which flows through the outlet while the pressed residue is discharged separately.



Specifications

- ❖ Type: Manual Operation
- ❖ Operation: Hand-operated flywheel with handle
- ❖ Drive System: Gear mechanism
- ❖ Pressing System: Screw shaft
- ❖ Feeding: Hopper
- ❖ Oil Output: Up to 0.7 L per batch
- ❖ Power Source: Manual operation — no electricity required
- ❖ Application: Oil extraction from oilseeds and nuts



**Technology University
(Mandalay)**
**Department of Mechanical
Engineering**

“Screw Press Oil Expeller”



Classification of Screw Press Oil Expeller

The following classification criteria help categorize manual screw press oil expeller based on their physical structure, capabilities, and intended applications.

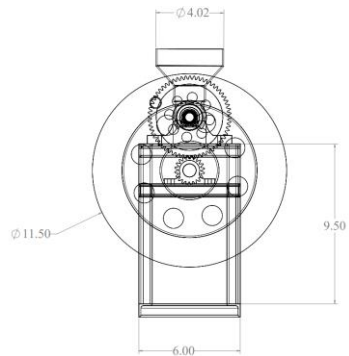
1. Configuration
2. Drive Mechanism
3. Pressing Chamber & Worm shaft

Type of Screw Press Oil Expeller

1. Manual Screw Press Oil Expeller
2. Motorized Screw Press Oil Expeller

Components of Manual Screw Press Oil Expeller

1. Flywheel
2. Gear & Pinion
3. Hopper
4. Screw Shaft
5. Barrel
6. Frame



Side View of Oil Expeller



Hopper



Gear and Pinion.



Barrel



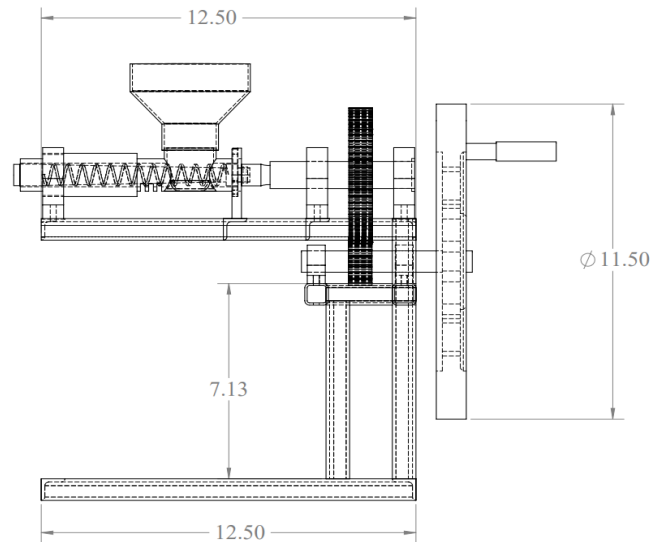
Screw Shaft.



Frame.



Flywheel

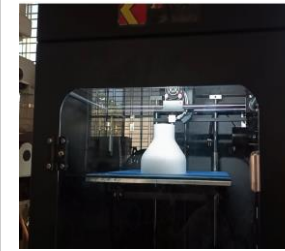


Front View of Oil Expeller



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Fabrication of Oil Expeller



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